SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

SYLLABUS - UDSC-301 - Statistics for Data Science

L – T – P – C	3 – 1 – 0 – 4	**Prerequisites**	Probability and Statistics, Calculus	Total Periods: **56**
Course Category	IDM			

Course Objectives

The main purpose of Statistical Inference is to make an inference about a population based on information contained in a random sample selected from that population, and to provide an associated measure of goodness for the inference. Another purpose is hypothesis testing for mean and variance of a normal distribution.

Course Outcomes

- Develop the skill set to estimate both point and interval estimations for a given parameter based on a sample from a normal population.
- Analyze the estimated parameter.
- Find an estimator for population proportion based on sample data.
- Develop testing hypotheses for a given real world problem.
- Apply different statistical tests based on data.
- Model the relationship between a scalar response and one or more explanatory variables.
- Find an appropriate regression model for a given data.

Syllabus

Unit	Topic	Contents	Periods
1	**Point Estimation**	Sampling Distributions and the Central Limit Theorem, Point Estimation: Unbiased Estimators, Variance of a Point Estimator, and Mean Squared Error of an Estimator, and Methods of Point Estimation: Method of Moments, Maximum Likelihood Estimators, Bayesian Estimation of Parameters	10
2	**Interval Estimation for a Single Sample**	Confidence Intervals on mean of a Normal Distribution, Confidence Interval on variance of a Normal Distribution, Large Sample Confidence Interval for a Population Proportion, Bootstrap Confidence Interval, Tolerance and Prediction Intervals	9
3	**Testing Hypotheses for a Single Sample**	Hypothesis Testing, Tests on the Mean and variance of a Normal Distribution, Tests on a Population Proportion, Testing for Goodness of Fit and Contingency Table Tests	10

Unit	Topic	Contents	Periods
4	**Statistical Inference for Two Samples**	Inference on the Difference in Means of two Normal Distributions, A Nonparametric Test for the Difference in Two Means, Paired t-test, Inference on the variance of Two Normal Distributions, Inference on Two Population Proportions	12
5	**Simple Linear Regression**	Simple Linear Regression, Least squares estimators, Hypothesis tests and Confidence Intervals in Single Linear Regression, Adequacy of the Regression Model: Residual Analysis, Coefficient of Determination, Regression on Transformed Variables, Logistic Regression, Polynomial Regression Models	15
		Total	**56**

Key Text(s)

- Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018. — Prescribed Chapters/Sections: Chapters 7–9 (up to 9.8), 10, 11.

Suggested Readings

- Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, Sixth Edition, Elsevier Academic Press, 2020.
- Irwin Miller, Marylees Miller, John E. Freund's Mathematical Statistics with Applications, Eighth Edition, Pearson Publications.